

BSCS FINAL PROJECT

Software Requirements Specification

LegacyLoop



Product Owner

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Presented by:

Group ID: xxxxxxxx

Student Reg#

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Faculty of Information Technology & Computer Science
University of Central Punjab

Software Requirements Specification

Version: #1

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S26CS043

Team

Member Name	Roles
M Awais Imran	Backend Development, API Engineering, Database Integration
Fatima shafeeq	Frontend Development (React Native), UI/UX Design
Alishba Niaz	AI Integration, Quiz & Assignment Generator, Chatbot Development

Table of Contents	i
Revision History & Sprint Log	ii
Previous Phases Feedback	ii
Abstract	iii
1. Introduction and Background	1
1.1 Product Vision & Problem Context	1
1.2 Stakeholders Identification	3
2. Agile Framework Setup	4
2.1 Epics For the Project	6
2.2 Product Backlog	7
2.3 User Stories	8
2.4 Story Points & Estimation	10
2.5 Sprint Planning & Sprint Backlog	11
3. Functional Requirements	12
3.1 Use Case Diagram	12
4. Non-functional Requirements	15
4.1 Reliability	15
4.2 Performance	15
4.3 Use-ability	15
4.4 Other Non-functional Requirements	16
5. Scrum Activities Plan	17
5.1 Kanban Board / Task Tracking	17
5.2 Acceptance Criteria	18
5.3 Release Criteria	20
5.4 Burndown Chart & Progress Tracking	21
5.5 Backlog Refinement Plan	22
6. Risk Analysis	23
7. References	24
Appendix A: Glossary	25
Appendix B: IV & V Report	26

Revision History & Sprint Log

Version	Sprint	Date	Changes	Owner

Previous Phases Feedback

S26 Idea Defence Feedback

Project Office













To:  L1S23BSCS0026 - ALISHBA NIAZ; +2 others

Wed 1/21/2026 7:29 PM

Cc:  Ihtisham Ul Haq

Dear Students,

Following is the feedback regarding your Idea Defence:

Sr. #	CS-25
Project Title	LEGACY LOOP

Similar products are already exist. May lead to plagiarism. Use is not well defined.

Scope is adequate. RAQ based external data source is to be processed very strictly. Misuse for plagiarism is very real possibility.

Very similar to an on-going FYP here in UCP. Feature set should be unique. Administrative activities should be there to monitor fake ads.

Your idea has been approved by the committee. Your next milestone is **Phase-1 (RS)**. You are advised to start working on it.

Abstract

Legacy Loop is a centralized academic ecosystem designed specifically for university students to improve peer-to-peer learning, resource sharing, and academic collaboration. Students often face difficulties accessing affordable study materials, verified mentorship, and organized academic discussions, as most interactions are scattered across informal social media groups. Legacy Loop addresses these challenges by providing a unified mobile platform where students can exchange academic resources, connect with mentors, participate in course-specific discussions, and utilize AI-powered academic assistance.

The platform integrates multiple modules including an ad-based marketplace for books and notes, a mentorship advertising hub, a tag-based academic forum, an AI chatbot, and a quiz & assignment generator. The AI modules utilize structured academic datasets derived from past papers and course materials to provide syllabus-aligned support and generate exam-oriented practice content.

Developed using React Native, ASP.NET Core, MongoDB Atlas, and SignalR, Legacy Loop formalizes campus academic interactions into a scalable mobile application. The platform aims to reduce educational costs, preserve institutional knowledge, improve exam preparation, and create a sustainable peer-learning environment where knowledge and resources are continuously shared between senior and junior students.

1. Introduction and Background

1.1 Product Vision & Problem Context

University students frequently face difficulties in accessing affordable study materials, verified mentorship, and organized academic discussions. Most academic interactions currently take place through scattered social media platforms such as WhatsApp and Facebook groups where information becomes difficult to manage, search, or preserve. Important academic resources such as textbooks, handwritten notes, and senior guidance are often lost after student's graduate, creating a gap between senior and junior students.

Legacy Loop is designed to solve this problem by creating a centralized academic ecosystem specifically tailored for university students. The platform provides a structured environment where students can exchange academic resources, connect with mentors, participate in course-specific discussions, and utilize AI-powered study assistance tools. By integrating all these features into a single mobile application, the system aims to improve academic collaboration and reduce dependency on fragmented communication channels.

The target users of the system include:

- Junior students seeking guidance, affordable resources, and exam preparation support
- Senior students offering mentorship, tutoring services, participating in academic discussions and forums, and educational materials
- System administrators responsible for moderation and management of the platform

The main objectives of Legacy Loop are:

- To centralize academic communication and resource sharing within the university ecosystem
- To reduce educational costs through the reuse and exchange of study materials
- To formalize peer-to-peer mentorship between senior and junior students
- To provide syllabus-aligned AI assistance through chatbot and quiz generation modules
- To preserve institutional knowledge using a searchable academic discussion forum
- To improve exam preparation through AI-generated quizzes and assignments.

Survey Method

A survey was conducted with 10 university students from different semesters and departments. The survey focused on identifying issues related to study material availability, mentorship accessibility, and academic communication.

Key findings from the survey included:

- Most students relied on WhatsApp groups for academic discussions
- Students found it difficult to search for old academic information in chats
- Many students considered textbooks and notes expensive
- Junior students struggled to find verified mentors for guidance
- Students preferred a single centralized academic platform

Interview Method

Interviews were conducted with 6 participants including junior students, senior students, and active student mentors. Participants highlighted the need for organized academic support, better communication channels, and easier access to university-specific study resources.

Observation of Current Process

The current academic collaboration process was observed through existing social media groups and informal peer interactions. It was identified that:

- Academic discussions are highly unorganized
- Valuable resources are often lost in chat histories
- No centralized mentorship system exists
- Generic AI tools do not provide university-specific assistance

Expected Benefits and Potential Beneficiaries

Legacy Loop is expected to provide several benefits to students and the university community, including:

- Easier access to affordable academic materials
- Improved collaboration between senior and junior students
- Faster and more organized academic communication
- Better exam preparation through AI-generated practice material
- Preservation of valuable academic knowledge within the university ecosystem

The primary beneficiaries of the system are university students who require academic support, mentorship, and affordable educational resources throughout their academic journey.

1.2 Stakeholders Identification

The following stakeholders are involved in or affected by the Legacy Loop system. Each stakeholder has specific roles, responsibilities, and expectations from the platform.

Stakeholder	Role in the System	Expectations
Junior Students	Primary users seeking academic guidance, study materials, and exam preparation support	Affordable resources, verified mentorship, organized academic discussions, and syllabus-aligned AI assistance
Senior Students	Mentors and resource providers offering tutoring services, books, notes, and academic guidance	Easy ad posting, communication tools, reputation building, and smooth interaction with juniors
Student Mentors	Provide tutoring and mentorship services through mentorship advertisements	Reliable communication system, visibility of mentorship profiles, and easy coordination with students
General University Students	Participate in academic forums and knowledge sharing activities	Access to organized discussions, searchable academic content, and peer collaboration
System Administrators	Manage platform operations, moderate content, and monitor system activities	Secure system management, content moderation tools, user management, and stable platform performance
Supervisor / Product Owner	Academic supervisor overseeing project development and progress	Successful implementation of project objectives, proper documentation, and efficient project workflow
Development Team	Design, develop, test, and maintain the system	Clear requirements, manageable Agile workflow, successful deployment, and maintainable architecture
University Community	Indirect beneficiaries of the academic ecosystem	Improved academic collaboration, reduced educational costs, and preservation of institutional knowledge

2. Agile Framework Setup

Legacy Loop will follow the Agile Scrum methodology to ensure iterative development, continuous feedback, and efficient collaboration among team members. The project will be divided into multiple sprints where each sprint focuses on delivering a specific set of features and improvements. Scrum practices will help the team manage tasks efficiently, adapt to changing requirements, and maintain steady development progress throughout the project lifecycle.

Team Roles

Role	Assigned Person	Responsibilities
Product Owner	Ihtisham Ul Haq	Define project vision, review progress, provide feedback, and approve sprint deliverables
Scrum Master	M Awais Imran	Manage sprint planning, coordinate team activities, remove workflow obstacles, and monitor Agile practices
Development Team	Fatima Shafeeq	Frontend development, React Native UI implementation, and user experience design
Development Team	Alishba Niaz	AI integration, chatbot implementation, and quiz & assignment generation features
Development Team	M Awais Imran	Backend development, API engineering, database integration, and real-time communication setup

Sprint Duration

The project will follow a sprint cycle of 2 weeks per sprint. Each sprint will include:

- Sprint Planning
- Task Assignment
- Development and Testing
- Sprint Review
- Sprint Retrospective

This approach ensures continuous delivery of functional modules and allows improvements based on feedback after every sprint.

Agile Tools and Technologies

The following tools will be used to support Agile development and collaboration:

Tool	Purpose
Git & GitHub	Version control and source code management
Trello	Sprint planning, Kanban board, and task tracking
Visual Studio	ASP.NET Core backend development
VS Code	React Native frontend development
MongoDB Atlas	Cloud-hosted database management
Postman	API testing and debugging
Figma	UI/UX wireframing and interface design

Workflow Approach

The team will follow a realistic Scrum workflow throughout the development process:

1. Product backlog items will be created based on system requirements and user needs.
2. During sprint planning meetings, selected backlog items will be moved into the sprint backlog.
3. Tasks will be distributed among team members according to their responsibilities.
4. Daily communication and progress tracking will be maintained using Trello and GitHub.
5. Completed features will be reviewed and tested at the end of every sprint.
6. Sprint retrospectives will be conducted to identify challenges and improve future development cycles.

This Agile workflow ensures better collaboration, continuous improvement, and timely delivery of project milestones.

2.1 Epics For the Project

The Legacy Loop system is divided into multiple high-level functional areas called Epics. Each epic represents a major module of the system and addresses a specific academic or operational problem within the university ecosystem.

Epic ID	Epic Name	Description	Related Problem Area
EP-01	User Management	Handles user registration, login, authentication, profile management, and role-based access control	Students unable to securely access and manage accounts
EP-02	Resource Marketplace	Allows students to create, manage, search, and browse ads for books, notes, and academic materials	Difficulty in accessing affordable academic resources
EP-03	Mentorship System	Enables senior students to post mentorship ads and connect with juniors for tutoring and guidance	Lack of organized peer-to-peer mentorship
EP-04	Academic Forum	Provides a tag-based discussion platform for posting academic questions and sharing knowledge	Academic discussions scattered across social media
EP-05	AI Chatbot	Assists students with academic queries using syllabus-aligned AI-generated responses	Generic AI tools provide irrelevant academic assistance
EP-06	Quiz & Assignment Generator	Generates quizzes, MCQs, and assignments using structured academic data and past paper patterns	Students lack exam-oriented practice material
EP-07	Real-Time Communication SignalR	Provides instant messaging and notifications using SignalR for smooth peer coordination	Delays and inefficiency in student communication
EP-08	Admin & Moderation System	Allows administrators to manage users, moderate content, and monitor platform activities	Lack of content control and platform monitoring
EP-09	Search & Filtering System	Enables efficient searching and filtering of ads, mentorships, and forum discussions using course tags	Difficulty finding relevant academic content quickly
EP-10	Notification System	Sends alerts and updates related to messages, mentorship replies, forum activities, and ads	Users may miss important academic interactions and updates

2.2 Product Backlog

The following product backlog contains the major features and tasks of Legacy Loop prioritized according to their importance and dependency within the system.

2.2.1 2.2 Product Backlog

Backlog ID	Epic	Feature Description	Priority
PB-01	User Management	User registration using email and password	High
PB-02	User Management	User login and logout functionality	High
PB-03	User Management	Update user profile information	Medium
PB-04	User Management	Upload and update profile picture	Medium
PB-05	User Management	Role-based access control for admin and students	Medium
PB-06	Resource Marketplace	Create resource advertisement for books and notes	High
PB-07	Resource Marketplace	Upload images for marketplace advertisements	High
PB-08	Resource Marketplace	Edit or delete marketplace advertisements	Medium
PB-09	Resource Marketplace	Search academic resources using course tags	High
PB-10	Resource Marketplace	Mark advertisements as sold or available	Medium
PB-11	Mentorship System	Create mentorship advertisement	High
PB-12	Mentorship System	Search mentors by subject expertise	High
PB-13	Mentorship System	View mentor academic profile and ratings	Medium
PB-14	Messaging System	Send real-time messages to mentors or sellers	High
PB-15	Messaging System	Receive real-time messages and notifications	High
PB-16	Messaging System	Generate meeting links for tutoring sessions	Medium
PB-17	Academic Forum	Post academic discussion posts	High
PB-18	Academic Forum	Reply to forum discussions	Medium
PB-19	Academic Forum	Mark helpful academic answers	Medium
PB-20	Academic Forum	Search forum discussions using course tags	Medium
PB-21	AI Chatbot	Ask academic questions through chatbot	High
PB-22	AI Chatbot	Receive syllabus-aligned AI responses	High
PB-23	Quiz & Assignment Generator	Generate MCQs from academic topics	High
PB-24	Quiz & Assignment Generator	Generate subjective questions and assignments	High
PB-25	Quiz & Assignment Generator	Select quiz difficulty level	Medium
PB-26	Notification System	Receive notifications for messages and replies	Medium
PB-27	Notification System	View personalized dashboard activities	Medium
PB-28	Dashboard System	Report reported posts and advertisements	Medium
PB-29	Admin & Moderation System	Manage users and platform content	Medium
	Admin & Moderation System		

2.3 User Stories

The following user stories are derived from the product backlog and describe the core functionalities of the Legacy Loop system from the user's perspective. These stories focus on user needs and expected outcomes to ensure the system delivers practical value to students, mentors, and administrators.

Story ID	Backlog ID	UserStory	Priority	Acceptance Criteria	Focal Person Responsible
US-01	PB-01	As a student, I want to register an account so that I can access the Legacy Loop platform securely.	High	User enters valid information → account created successfully. Invalid data → error message displayed.	Awais
US-02	PB-02	As a student, I want to log in to my account so that I can access the marketplace, mentorship, and AI features.	High	Valid credentials → login success. Invalid credentials → access denied message shown.	Awais
US-03	PB-03	As a student, I want to create marketplace ads so that I can sell or donate academic resources.	High	User uploads valid ad details and image → ad published successfully.	Fatima
US-04	PB-04	As a student, I want to search and filter resources so that I can quickly find relevant books or notes.	High	Search results displayed according to course tags and keywords.	Fatima
US-05	PB-05	As a junior student, I want to find mentors so that I can get academic guidance for specific subjects.	High	Mentor listings displayed with subject expertise and ratings.	Alishba
US-06	PB-06	As a senior student, I want to post mentorship advertisements so that juniors can contact me for tutoring.	High	Mentorship ads are created and visible to other students.	Alishba
US-07	PB-07	As a student, I want to chat with mentors or sellers so that I can coordinate sessions or purchases easily.	High	Messages are delivered instantly through real-time chat.	Awais
US-08	PB-08	As a student, I want to use the AI chatbot so that I can get syllabus-related academic assistance.	High	User enters query → chatbot returns relevant academic response.	Alishba

Legacy Loop

US-09	PB-09	As a student, I want to generate quizzes and assignments so that I can prepare effectively for exams.	High	System generates questions based on selected courses and difficulty.	Alishba
US-10	PB-10	As a student, I want to participate in discussion forums so that I can ask questions and share knowledge with my peers.	Medium	Users can create posts, answer questions, and upvote responses.	Fatima
US-11	PB-11	As a student, I want to manage my academic profile so that I can update my expertise and active listings.	Medium	Updated profile information was saved successfully.	Fatima
US-12	PB-12	As an administrator, I want moderate content so that inappropriate or spam content can be removed.	Medium	Admin can remove flagged ads/posts successfully.	Awais

2.4 Story Points & Estimation

The following table presents the estimated effort required for implementing each user story in Legacy Loop. Story points are assigned based on the complexity of the feature, development time, integration requirements, and technical dependencies.

Story ID	User Story (Short)	Complexity Level	Story Points	Justification
US-01	User Registration	Medium	3	Requires form validation, database storage, and error handling
US-02	User Login	Medium	3	Authentication logic and JWT session handling required
US-03	Create Marketplace Ads	High	5	Includes image upload, ad management, and MongoDB integration
US-04	Search & Filter Resources	Medium	3	Requires indexed search and filtering by course tags
US-05	Find Mentors	Medium	3	Requires mentor listing, filtering, and profile display
US-06	Post Mentorship Ads	Medium	3	Includes ad creation and mentorship management features
US-07	Real-Time Chat	High	8	SignalR integration and instant messaging functionality required
US-08	AI Chatbot	High	8	Requires AI API integration and syllabus-aware response generation
US-09	Quiz & Assignment Generator	High	8	Complex AI-generated content creation with difficulty selection
US-10	Discussion Forum	Medium	5	Includes post creation, replies, tagging, and upvote system
US-11	Manage Academic Profile	Low	2	Simple profile update and listing management functionality
US-12	Content Moderation	Medium	5	Requires admin controls, reporting system, and moderation logic

2.5 Sprint Planning & Sprint Backlog

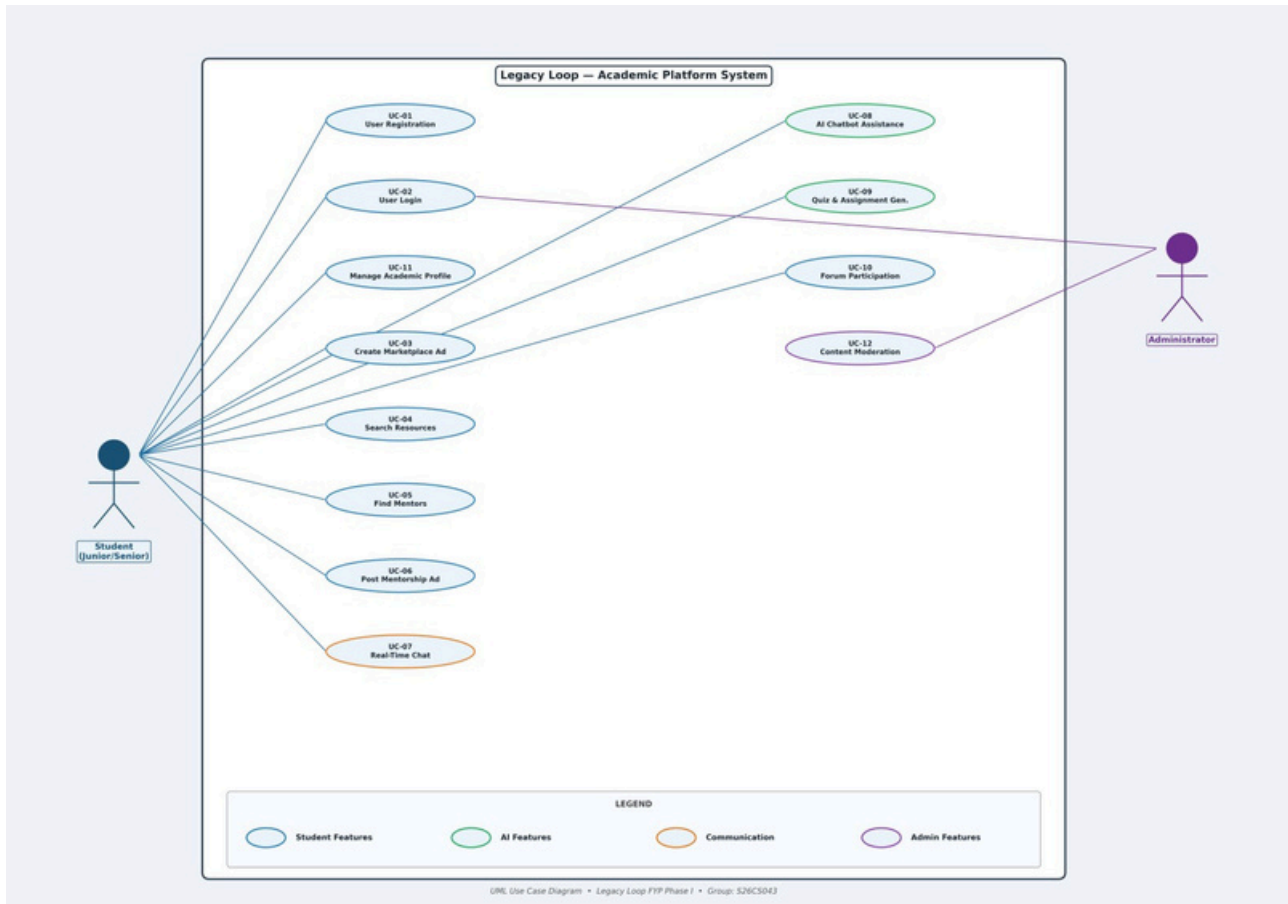
The development of Legacy Loop will follow an Agile Scrum methodology where project tasks are divided into multiple sprints. Each sprint focuses on a specific set of user stories and project goals to ensure steady progress and manageable workload distribution. The sprint backlog is organized according to feature dependencies, complexity levels, and overall system priorities, allowing the team to incrementally develop and test core functionalities of the platform.

Sprint No.	Duration	Selected User Stories	Sprint Goal	Total Story Points
Sprint 1	2 Weeks	US-01, US-02, US-11	Implement user authentication and profile management features	8
Sprint 2	2 Weeks	US-03, US-04, US-06	Develop marketplace and mentorship advertisement modules	11
Sprint 3	2 Weeks	US-05, US-07	Implement mentor discovery and real-time communication system	11
Sprint 4	2 Weeks	US-08, US-09	Integrate AI chatbot and quiz & assignment generator	16
Sprint 5	2 Weeks	US-10, US-12	Develop discussion forum and admin moderation functionalities	10

3. Functional Requirements

The functional requirements of Legacy Loop are derived from the defined user stories and describe the major interactions between users and the system. These requirements are represented using use cases to clearly identify the actions performed by different actors such as students, mentors, and administrators. The use case model helps in understanding how users interact with the platform to perform tasks such as authentication, resource sharing, mentorship coordination, AI-assisted learning, and forum participation.

3.1 Use Case Diagram



3.2 Use Case Description Table

Use Case ID	Use Case Name	Primary Actor	Description	Mapped User Story ID	User Story
UC-01	User Registration	Student	Allows a new user to create an account using valid credentials	US-01	As a student, I want to register an account so that I can access the Legacy Loop platform securely
UC-02	User Login	Student	Enables users to securely log into the system	US-02	As a student, I want to log in to my account so that I can access marketplace, mentorship, and AI features
UC-03	Create Marketplace Advertisement	Student	Allows users to create ads for books, notes, and academic resources	US-03	As a student, I want to create marketplace ads so that I can sell or donate academic resources
UC-04	Search Resources	Student	Enables users to search and filter academic resources	US-04	As a student, I want to search and filter resources so that I can quickly find relevant books or notes
UC-05	Find Mentors	Junior Student	Allows students to browse and find mentors for academic guidance	US-05	As a junior student, I want to find mentors so that I can get academic guidance for specific subjects
UC-06	Post Mentorship Advertisement	Senior Student	Allows mentors to post tutoring and mentorship advertisements	US-06	As a senior student, I want to post mentorship advertisements so that juniors can contact me for tutoring
UC-07	Real-Time Chat	Student	Enables instant communication between mentors, buyers, and students	US-07	As a student, I want to chat with mentors or sellers so that I can coordinate sessions or purchases easily

Legacy Loop

UC-08	AI Chatbot Assistance	Student	Provides syllabus-related academic assistance through AI chatbot integration	US-08	As a student, I want to use the AI chatbot so that I can get syllabus-related academic assistance
UC-09	Quiz & Assignment Generation	Student	Generates quizzes and assignments based on selected topics and difficulty	US-09	As a student, I want to generate quizzes and assignments so that I can prepare effectively for exams
UC-10	Forum Participation	Student	Allows users to create posts, answer questions, and participate in discussions	US-10	As a student, I want to participate in discussion forums so that I can ask questions and share knowledge with peers
UC-11	Manage Academic Profile	Student	Enables users to update profile details and manage listings	US-11	As a student, I want to manage my academic profile so that I can update my expertise and active listings
UC-12	Content Moderation	Administrator	Allows administrators to monitor and remove inappropriate content	US-12	As an administrator, I want to moderate content so that inappropriate or spam content can be removed

4. Non-functional Requirements

The non-functional requirements of Legacy Loop define the quality attributes and operational standards that the system must maintain to ensure reliability, efficiency, usability, and security. These requirements focus on how the system performs under different conditions and ensure that users receive a stable, secure, and user-friendly experience while using the platform.

4.1 Reliability

Requirement ID	Requirement Description
REL-01	The system shall maintain stable performance during peak academic periods such as midterms and finals.
REL-02	The platform shall ensure that user data is securely stored and retrievable without corruption or loss.
REL-03	The system shall automatically reconnect real-time chat sessions in case of temporary network interruption.
REL-04	The application shall provide backup and recovery mechanisms for critical user and marketplace data.
REL-05	The platform shall remain operational even if certain external AI services become temporarily unavailable.

4.2 Performance

Requirement ID	Requirement Description
PER-01	User authentication requests shall be processed within a few seconds under normal load conditions.
PER-02	Marketplace search results shall be displayed within 2–3 seconds.
PER-03	Real-time chat messages shall be delivered instantly using SignalR technology.
PER-04	The AI chatbot shall generate responses within an acceptable response time for academic queries. I.e. 15 sec
PER-05	The mobile application shall maintain smooth navigation and responsive user interaction across supported devices.

4.3 Usability

Requirement ID	Requirement Description
USE-01	The system shall provide a simple and user-friendly mobile interface for students.
USE-02	Users shall be able to navigate marketplace, mentorship, and forum features without technical assistance.
USE-03	The platform shall provide clear validation and error messages for invalid user actions.
USE-04	The application shall support intuitive search and filtering options for academic resources.
USE-05	The system shall use terminology familiar to university students to improve usability and accessibility.

4.4 Other Non-functional Requirements

Requirement ID	Requirement Description
ONFR-01	All communication between the mobile application and backend services shall be secured using HTTPS/TLS for encryption.
ONFR-02	The system shall implement JWT-based authentication and role-based access control mechanisms.
ONFR-03	The platform shall support both Android and iOS devices using React Native.
ONFR-04	The system shall be scalable enough to support increasing numbers of users and academic content.
ONFR-05	The platform shall maintain compatibility with cloud-hosted MongoDB database infrastructure.
ONFR-06	The application shall include moderation and reporting features to prevent inappropriate content sharing.

5. Scrum Activities Plan

The Legacy Loop project will follow the Scrum Agile methodology to ensure continuous development, collaboration, and iterative improvement throughout the project lifecycle. Daily Scrum meetings will be conducted for 10–15 minutes to discuss completed work, current progress, and blockers faced by team members. At the end of each sprint, a Sprint Review meeting will be held to demonstrate completed features and gather feedback from the supervisor and team members. Sprint Retrospective sessions will also be conducted after every sprint to evaluate team performance, identify challenges, and improve the development workflow for upcoming sprints.

5.1 Kanban Board / Task Tracking

To ensure effective task management and continuous visibility of project progress, the Legacy Loop development team will use a Kanban-based workflow methodology. The team will manage project activities using Trello/Jira boards where tasks will be categorized into different stages such as To Do, In Progress, Testing, and Done. Each task will be assigned to a responsible team member along with deadlines and priority levels. This approach will help the team monitor sprint progress, identify bottlenecks, maintain workload balance, and ensure timely completion of project deliverables. The Kanban workflow for Legacy Loop is illustrated below:

To Do	In Progress	Testing	Done
Design user registration UI	Implement JWT authentication API	Test login validation and error handling	Setup React Native project structure
Design marketplace interface	Develop real-time chat using SignalR	Validate marketplace advertisement posting	Configure MongoDB database
Create mentorship module UI	Integrate AI chatbot API	Test AI-generated quiz functionality	Complete database schema design
Design discussion forum pages	Implement resource search and filtering	Perform security and authentication testing	Authentication module completed
Create admin dashboard layout	Develop forum posting and commenting system	Validate profile management functionality	Initial UI wireframes approved
Define course tagging structure	Implement mentorship advertisement module	Test mobile responsiveness on Android/iOS	API routing and backend setup completed
Create notification system UI	Develop quiz and assignment generator	Test real-time messaging delivery	Basic navigation and routing completed
Prepare reporting and moderation screens	Implement role-based access control	Validate content moderation functionality	Cloud database connection configured
Design profile management screens	Integrate push notifications	Test discussion forum interactions	Sprint 1 deliverables completed
Prepare deployment configuration	Optimize backend API performance	Conduct final integration testing	Initial testing environment setup completed

5.2 Acceptance Criteria

The acceptance criteria for Legacy Loop define the conditions that must be satisfied for each feature to be considered successfully implemented and ready for deployment. These criteria are based on the defined user stories and functional requirements of the system. Each feature will be tested through specific test scenarios to verify correct functionality, expected behavior, usability, and system reliability.

Feature	Test Scenario	Expected Result
User Registration	User enters valid registration details and submits form	Account created successfully and user redirected to login screen
User Login	User enters valid email and password	User authenticated successfully and dashboard displayed
Invalid Login Handling	User enters incorrect credentials	Error message displayed and login denied
Marketplace Advertisement	User uploads book/resource details with image	Advertisement published successfully in marketplace
Resource Search & Filtering	User searches resources using keywords or course tags	Relevant academic resources displayed correctly
Mentorship Advertisement	Mentor creates tutoring advertisement	Advertisement becomes visible to students
Mentor Discovery	Student searches available mentors	Mentor profiles displayed based on selected criteria
Real-Time Chat	User sends message to another user	Message delivered instantly without refresh
AI Chatbot Assistance	User submits syllabus-related academic query	AI chatbot returns relevant and meaningful response
Quiz & Assignment Generator	User selects course and difficulty level	Quiz or assignment generated successfully
Discussion Forum Posting	User creates forum discussion post	Post published successfully in forum section
Forum Interaction	User comments or replies to discussion	Response displayed immediately in discussion thread
Profile Management	User updates profile information	Updated profile information saved successfully
Content Moderation	Admin removes inappropriate content	Selected content deleted from system successfully
Push Notifications	System sends new message or update notification	User receives notification instantly

Legacy Loop

Mobile Responsiveness	User accesses system on different mobile devices	Interface remains responsive and usable
Security Validation	Unauthorized user attempts protected action	Access denied and security warning displayed
System Stability	Multiple users access the platform simultaneously	System performs without crashes or major delays

The above acceptance criteria ensure that all major functionalities of Legacy Loop operate according to the defined user requirements and project objectives. Successful completion of these test scenarios will confirm that the system is reliable, secure, user-friendly, and ready for deployment in a real academic environment.

5.3 Release Criteria

The Legacy Loop application will be considered ready for release once all critical functional and non-functional requirements have been successfully implemented, tested, and approved. The release criteria ensure that the system is stable, secure, and capable of delivering the expected user experience before deployment.

Release Criteria ID	Release Condition	Expected Outcome
RC-01	All high-priority user stories are completed	Core system features function correctly
RC-02	User authentication and authorization modules pass testing	Secure login and access control verified
RC-03	Marketplace, mentorship, and discussion forum modules operate without critical issues	Users can interact with all major platform features successfully
RC-04	AI chatbot and quiz generation modules produce valid academic responses	AI-assisted learning features function correctly
RC-05	Real-time chat and notification systems perform successfully	Users receive instant communication updates
RC-06	All identified critical and high-severity bugs are resolved	System stability and reliability ensured
RC-07	System passes security and validation testing	Unauthorized access and data vulnerabilities prevented
RC-08	Application performs correctly on supported Android and iOS devices	Mobile responsiveness and compatibility confirmed
RC-09	Final sprint review and acceptance testing are completed	Development objectives verified successfully
RC-10	Project supervisor approves final system demonstration	System officially approved for release

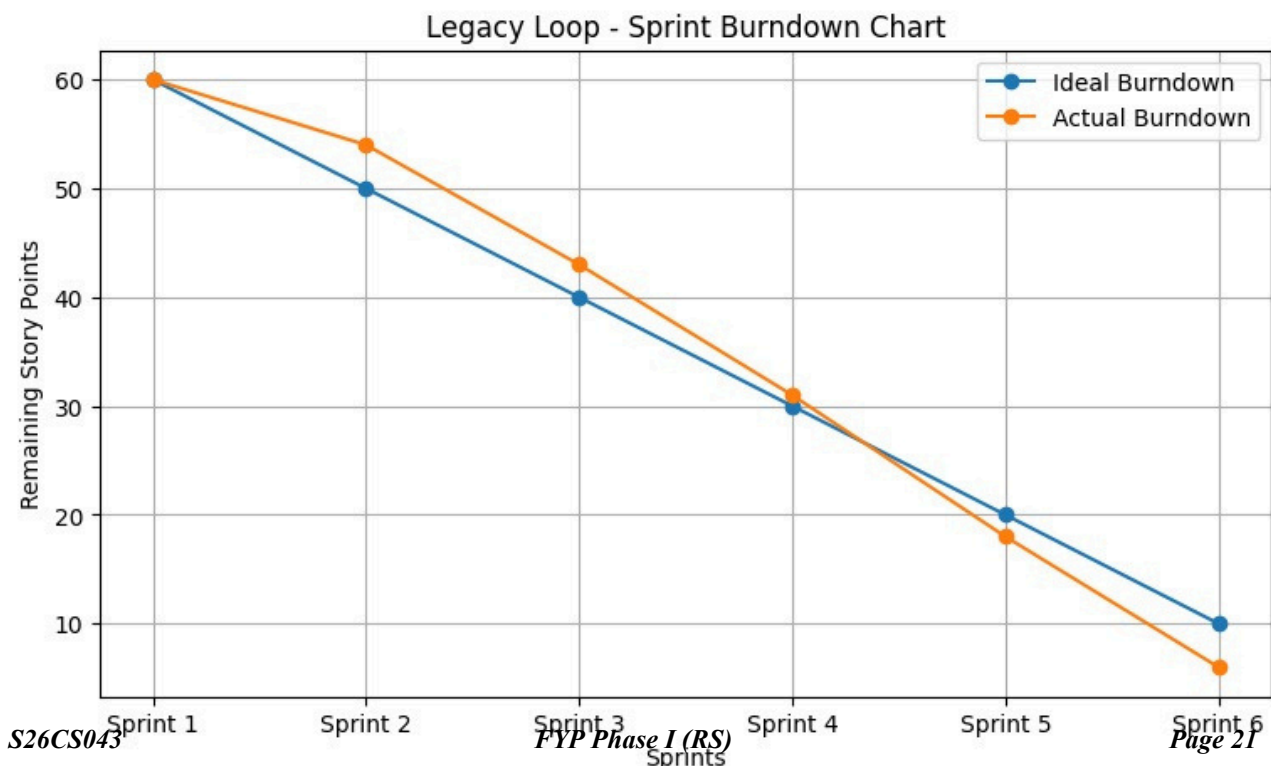
The above release criteria ensure that Legacy Loop meets the required quality standards, functional expectations, and security requirements before deployment. Fulfillment of these conditions confirms that the system is reliable, fully tested, and ready for practical use in an academic environment.

5.4 Burndown Chart & Progress Tracking

The progress of the Legacy Loop project will be monitored using sprint burndown charts and Agile task tracking techniques. The development team will track completed and remaining work during each sprint using story points assigned to user stories. Project tasks and sprint activities will be managed through tools such as Trello or Jira.

The burndown chart will represent the relationship between remaining work and sprint duration. As sprint tasks are completed, the remaining story points are expected to decrease gradually throughout the sprint. Daily Scrum meetings will be used to update task progress. Any unexpected workload increases will be analyzed during sprint reviews to improve team productivity.

Sprint	Total Story Points	Expected Burn Rate	Progress Tracking Method
Sprint 1	8	1–2 story points completed per day	Daily Scrum meetings and Trello/Jira updates
Sprint 2	11	1–2 story points completed per day	Sprint backlog tracking and Kanban board monitoring
Sprint 3	11	1–2 story points completed per day	Task completion reports and sprint reviews
Sprint 4	10	2–3 story points completed per day	Task completion reports and sprint reviews
Sprint 5	10	1–2 story points completed per day	Burndown chart analysis and daily progress updates
Sprint 6	6	1–2 story points completed per day	Final sprint review and project milestone tracking



5.5 Backlog Refinement Plan

The product backlog for Legacy Loop will be continuously reviewed and refined throughout the project development lifecycle to ensure that the project requirements remain clear, prioritized, and aligned with stakeholder expectations. Backlog refinement activities will help the team update user stories, improve task estimates, identify new requirements, and manage changing project needs effectively.

Refinement Aspect	Description
Frequency	Backlog refinement sessions will be conducted at the end of every sprint and before the start of the next sprint
Participants	All development team members and the project supervisor will participate in refinement discussions
Requirement Updates	New features, improvements, or change requests will be added to the product backlog when required
Priority Adjustment	User stories and tasks will be reprioritized based on project goals, complexity, and stakeholder feedback
Story Clarification	Existing user stories will be updated to improve clarity, acceptance criteria, and implementation details
Estimation Review	Story points and workload estimates will be reviewed and adjusted according to project progress
Dependency Identification	Technical and functional dependencies between tasks will be analyzed during refinement sessions
Risk Discussion	Potential development risks and blockers will be identified and discussed proactively
Sprint Preparation	Refined backlog items will be prepared and finalized before sprint planning meetings
Continuous Improvement	Feedback from sprint reviews and retrospectives will be used to improve backlog quality and team workflow

The backlog refinement process will support continuous improvement within the Legacy Loop project by ensuring that development priorities remain organized, achievable, and responsive to project changes. Regular refinement sessions will help the team maintain an efficient workflow, reduce ambiguities, and improve sprint planning accuracy throughout the Agile development process.

6. Risk Analysis

Risk analysis is performed to identify potential technical, and team-related challenges that may affect the successful development of the Legacy Loop project. Identifying risks at an early stage allows the development team to prepare mitigation strategies and reduce the possibility of project delays, system failures, or performance issues during the Agile development lifecycle.

Risk ID	Risk Description	Impact	Mitigation Strategy
R-01	Delay in AI chatbot and quiz generator integration	High	Use well-documented AI APIs and allocate additional testing time
R-02	Real-time chat functionality may experience synchronization issues	High	Conduct continuous SignalR testing and optimize server communication
R-03	Team members may face workload imbalance during sprints	Medium	Distribute tasks evenly and monitor sprint workload regularly
R-04	Changes in project requirements during development	Medium	Perform regular backlog refinement and sprint review meetings
R-05	Database performance issues with increasing user data	High	Optimize MongoDB queries and implement indexing techniques
R-06	Security vulnerabilities in authentication and APIs	High	Implement JWT security, input validation, and regular security testing
R-07	Limited experience with certain technologies or frameworks	Medium	Conduct self-learning sessions and technical research before implementation
R-08	Integration conflicts between frontend, backend, and AI modules	Medium	Follow modular development and perform integration testing frequently
R-09	Mobile responsiveness issues across different devices	Medium	Test the application regularly on Android and iOS devices
R-10	Internet or cloud service downtime affecting system availability	Medium	Use backup services and maintain proper error-handling mechanisms
R-11	Delays in task completion during sprint execution	Medium	Track daily progress through Scrum meetings and adjust sprint tasks if necessary
R-12	Loss of project data or configuration files	High	Maintain regular backups using GitHub and cloud storage repositories

7. References

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Appendix A: Glossary

Term / Acronym	Definition
Legacy Loop	The proposed academic support and resource sharing platform developed for university students
SRS	Software Requirements Specification document that defines system requirements and functionalities
Agile Scrum	An Agile software development methodology based on iterative sprint-based development
Sprint	A fixed development cycle in Scrum used to complete a set of project tasks
Product Backlog User	A prioritized list of project requirements, features, and user stories
Story	A short description of a system feature written from the user's perspective
Use Case	A description of interactions between a user and the system to achieve a goal
API	Application Programming Interface used for communication between software components
JWT	JSON Web Token used for secure authentication and authorization
MongoDB	A NoSQL database management system used for storing project data
React Native	A cross-platform mobile application development framework
.NET	A software development framework used for backend development
SignalR	A real-time communication library used for instant messaging and notifications
AI	Artificial Intelligence technology used for chatbot and quiz generation features
UI	User Interface of the mobile application
UX	User Experience related to usability and interaction design
CRUD Operations	Create, Read, Update, and Delete operations performed on system data
Authentication	The process of verifying a user's identity before granting system access
Authorization	The process of controlling user permissions and access levels
Cloud Database Push Notification	A remotely hosted database accessible through internet services
Kanban Board	A real-time alert or message sent to users through the application
Burndown Chart	A task-tracking board used to monitor project workflow and progress
Mentorship	A Scrum chart used to track remaining work during a sprint
Marketplace Module	A system component that connects junior students with mentors
Discussion Module	A feature that allows students to buy, sell, or share academic resources
Forum	A collaborative platform where users can post questions and share discussions

Appendix B: IV & V Report
(Independent verification & validation)
IV & V Resource

Name _____ Signature _____

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
...					

Table : List of non-trivial defects